

Problem 89 (Tournament of Towns 2013 Junior A-Level Problem 1). There are n positive integers such that the sum of any two of them (the two integers should be distinct) is a power of 2. Find the largest possible value of n .

Solution. The answer is 2. Clearly, there can be 2 numbers (e.g. 1 and 3). We now assume $n \geq 2$. Let a be the largest number, and let b be any other number. Suppose $2^m \leq a < 2^{m+1}$. Then we have $2^m < a + b < 2a < 2^{m+2}$. This implies $a + b = 2^{m+1}$, and hence b is uniquely determined. This shows n is at most 2. $\square$

Problem 90 (International Zhautykov Olympiad 2022 Problem 1). Let $P(x), Q(x)$ and $R(x)$ be nonzero polynomials with real coefficients such that

$$P(x) + Q(x) + R(x) = P(Q(x)) + Q(R(x)) + R(P(x)) = 0.$$

Prove that the degrees of all these polynomials are even.

Solution. Let n be the largest of the degree of the three polynomials. Denote by a, b and c the coefficients of x^n at $P(x), Q(x)$ and $R(x)$ respectively (some of those coefficients may vanish).

The coefficients of x^n at $P(x) + Q(x) + R(x)$, as well as of x^{n^2} at $P(Q(x)) + Q(R(x)) + R(P(x))$, both vanish. Hence,

$$a + b + c = 0 \quad \text{and} \quad ab^n + bc^n + ca^n = 0. \quad (1)$$

The first equality yields that at least two numbers among a, b and c are nonzero. If the third number (say, c) vanishes, then the second equality is violated. Therefore, all three polynomials have degree n , and we need to prove n is even.

Assume the contrary, for the sake of contradiction. WLOG, the numbers a and b have the same sign. Changing the sign of all three numbers a, b and c , if necessary, we achieve $a, b > 0$ (where (1) remains unchanged). Then we have $c = -(a + b) < 0$ and $0 < a, b < |c|$; hence bc^n and ca^n are negative, and therefore

$$|bc^n + ca^n| > |bc^n| = |c| \cdot b \cdot |c|^{n-1} > a \cdot b \cdot b^{n-1} = ab^n.$$

This contradicts the second equality in (1).

Thus, all three degrees are equal to an even number n . $\square$

Problem 91 (International Zhautykov Olympiad 2022 Problem 3). Let $ABCD$ be a parallelogram where $\angle A$ is an acute angle. Let E be a point inside the parallelogram such that $AB = BE = CE$. The extension of the line CE meets the segment AD at F . Reflect the point F in the line DE to obtain the point F' . The line EF' meets the segment AD at P . Let Q be an intersection point of the circumcircles of $\triangle ADE$ and $\triangle CFF'$ such that A and Q lie on the same side of the line EF' . Prove that $\angle CQE = \angle DQP$.

Solution. Firstly, $\triangle CDE$ is isosceles since $CD = AB = CE$. As

$$\angle F'ED = \angle FED = 180^\circ - \angle CED = 180^\circ - \angle CDE,$$

we have $F'E \parallel CD$. Let C' be the reflection of C in the line DE . Then C' lies on (CFF') and $CDC'E$ is a rhombus. As $F'E \parallel CD$ and $EC' \parallel CD$, the point C' lies on $F'E$. Also, $C'ABE$ is a rhombus because it is a parallelogram with $AB = BE$. Now, since $C'A = C'D$, we have

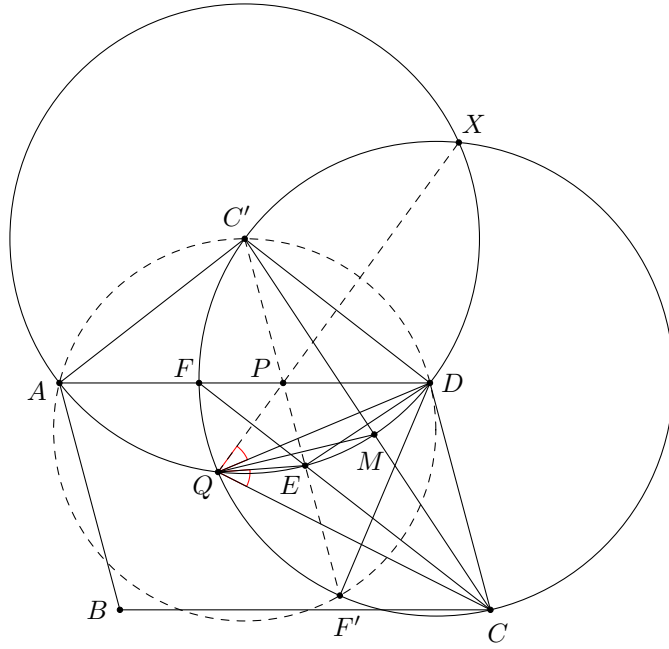
$$\angle C'AD = \angle C'DA = \angle CFD = \angle EFD = \angle EF'D.$$

This implies C', A, F', D are concyclic. It follows that $P = AD \cap C'F'$ lies on the radical axis of (ADE) and (CFF') .

Let X be the second intersection point of (ADE) and (CFF') , and let M be the intersection point of the segment CC' and (ADE) . As $MD = ME$, the line QM is an angle bisector of $\angle DQE$. Noting C' is the centre of (ADE) (since $C'A = C'D = C'E$), we find that

$$\angle MQX = \frac{1}{2} \angle MC'X = \frac{1}{2} \angle CQX.$$

This shows QM is also an angle bisector of $\angle CQX$. It follows easily that $\angle CQE = \angle DQP$. $\square$



Problem 92 (RMM 2015 Problem 3). Let n be an integer greater than 100. Prove that there are finitely many nonnegative integers k having the following property.

Property. Initially, we write down the integers $k+1, k+2, \dots, k+n$ on the blackboard. In each step, we choose any two numbers a and b on the blackboard and replace them by one of the following numbers.

$$a+b, a-b, b-a, ab, \frac{a}{b}, \frac{b}{a}$$

(We can only do division if the denominator is nonzero.) After $n-1$ steps, it is possible to obtain the single number $n!$.

Solution. Suppose on the contrary that there are infinitely many such k having the property. Note that there is a finite number of sequences of operations that reduce the n integers on the blackboard to a single number. Thus, there are infinitely many such k for which $n!$ can be obtained from the integers $k+1, k+2, \dots, k+n$ using exactly the same sequence of operations. If we regard k as a variable, and regard $k+1, k+2, \dots, k+n$ as polynomials in k , the same sequence of operations always reduces the numbers to a rational function $f(k)$ in the end. As $f(k) = n!$ for infinitely many k , the rational function $f(k)$ is exactly the constant polynomial $n!$.

For any polynomial $P(x)$, let $g(P)$ be the sum of the absolute values of all its coefficients. Then we have $g(P+Q) \leq g(P) + g(Q)$ and $g(PQ) \leq g(P)g(Q)$. Also, for a rational function $\frac{P(x)}{Q(x)}$, define $g\left(\frac{P}{Q}\right) = g(P) + g(Q)$. In the following, we do not cancel any common factors in the numerator and the denominator (for example, $g\left(\frac{x^2}{x}\right) = g(x^2) + g(x) = 1 + 1 = 2$ is different from $g(x) = 1$).

We claim that $X = \prod_r g(r)$ is monotonic decreasing where r runs through all rational functions on the blackboard. It suffices to prove that this quantity does not increase in each operation. Indeed, we check that

$$\begin{aligned} g\left(\frac{P}{Q} \pm \frac{R}{S}\right) &= g\left(\frac{PS \pm QR}{QS}\right) \leq g(PS \pm QR) + g(QS) \leq g(P)g(S) + g(Q)g(R) + g(Q)g(S) \leq g\left(\frac{P}{Q}\right)g\left(\frac{R}{S}\right), \\ g\left(\frac{P}{Q} \cdot \frac{R}{S}\right) &= g\left(\frac{PR}{QS}\right) = g(PR) + g(QS) \leq g(P)g(R) + g(Q)g(S) \leq g\left(\frac{P}{Q}\right)g\left(\frac{R}{S}\right), \\ g\left(\frac{P}{Q} \div \frac{R}{S}\right) &= g\left(\frac{PS}{QR}\right) = g(PS) + g(QR) \leq g(P)g(S) + g(Q)g(R) \leq g\left(\frac{P}{Q}\right)g\left(\frac{R}{S}\right). \end{aligned}$$

So our claim holds.

On the other hand, we replace k by $k-m$ where $m = \left\lfloor \frac{n}{2} \right\rfloor$ so that the initial numbers are $k-m+1, k-m+2, \dots, k-m+n$. It is still true that there are infinitely many k for which the final number obtained is $n!$ under the above sequence of operations. However, the initial value of X is

$$m(m-1) \cdots (2)(1)(2) \cdots (n-m+1) \leq m!(m+2)! < n!$$

for $n > 7$ (where the last inequality can be easily proved by induction). This is a contradiction as the final value of X is at least $n!$ (it needs not be exactly $n!$ as we do not cancel any common factors in the intermediate steps). $\square$